MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FOURTH SEMESTER EXAMINATION

(2024 Admission Onwards)

MG4DSCCSC200 – OOPs Concepts using JAVA

Duration: 1.5 hours Maximum Marks: 50

Students should attempt at least one question from each Course Outcome (CO) to enhance their overall outcome attainability

Part A

Very Short Answer Questions

(Answer **all** questions, each question carries **2** marks.)

1. State the role of Java Virtual Machine. [U][CO1]
2. Identify the use of the super keyword. [U][CO2]
3. Analyze how dynamic method dispatch supports runtime polymorphism. [An][CO2]
4. Define exception handling. [K][CO3]
5. Apply multithreading to execute two independent tasks simultaneously in Java. [A][CO3]

[5 × 2 = 10]

Part B

Short Answer Questions

(Answer **5** out of **7** questions, each question carries **5** marks.)

6. Explain the concepts and benefits of Object Oriented Programming. [U][CO1]
7. Apply different types of Java operators to evaluate an arithmetic and logical expression.[A][CO1]
8. Describe the structure of a Java class and its components. [U][CO2]
9. Illustrate how Java resolves method overloading at compile time by analyzing a suitable program. [An][CO2]
10. Use inheritance and method overriding in Java to demonstrate runtime polymorphism with a suitable program. [A][CO2]
11. Discuss Java API packages and their significance. [U][CO3]
12. Write a Java program to create multiple threads and demonstrate the thread life cycle.[A][CO3]

[5 × 5 = 25]

Part C

Essay Questions

(Answer **1** out of **2** questions, each question carries **15** marks.)

13. Explain Java control statements with suitable programs and examples. [U][CO1]
14. Explain Swing components and event handling mechanism using the Delegation Event Model. [A][CO3]

[1 × 15 = 15]